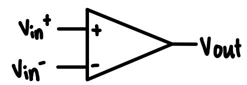
Op-amp devices and Digital Logic



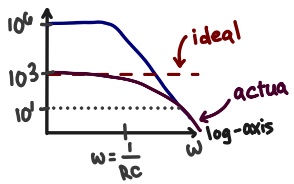
$V_{out} = A_{OL}(V_{in}^+ - V_{in}^-)$

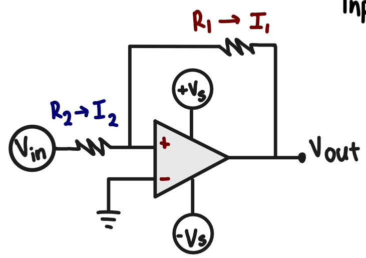
$A_{OL} \gg 1$

Golden Rule

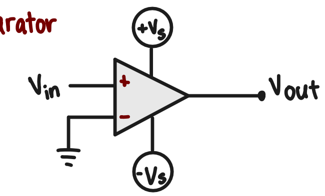
$V_{in}^+ = V_{in}^- = 0 \qquad I_1 = I_2$

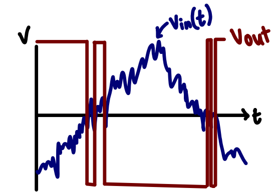
Input draws no current



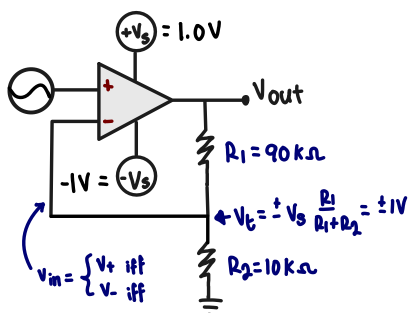


Comparator

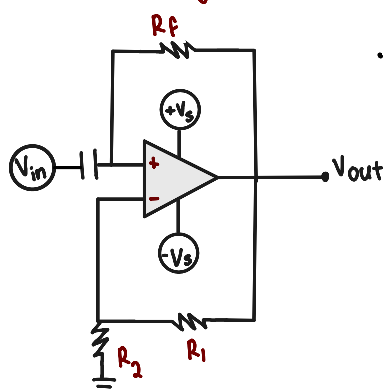




Schmitt Trigger - Reduces Switching Behavior

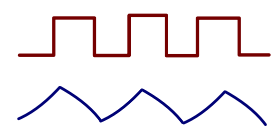


Relaxing Oscillator

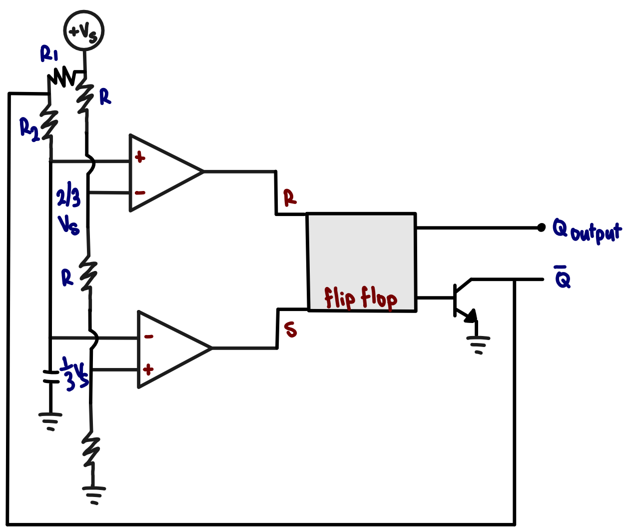


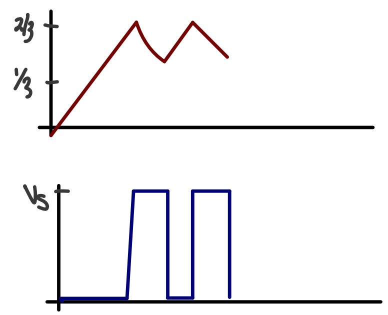
Assuming Starting Point

- $V_{out} = +V_s$ + C uncharged
- $V_{in}^- = 0 < V_{in}^+ = V_+ = \frac{R_2}{R_1+R_2} \cdot V_s$
- since $V_{out} = V_s$ a current will flow through R_f + start to charge C with $\tau = R_f C$
- eventually $V_{in}^- > V_{in}^+ = V_+$
- then $V_{out} \to -V_s$
- Cap starts to discharge
- then output flips back



555 Timer Integrated Circuit





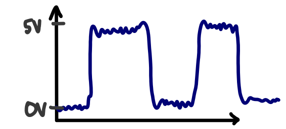
Digital Logic

- a 'bit' of information - 0 or 1 state

⇕

±0V or +5V

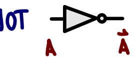
(0-3V) (3-5V)



Logic Gate

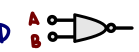
TRUTH TABLE

NOT



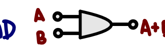
Input	output
0	1
1	0

NAND



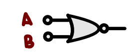
Input A	Input B	output
0	0	1
1	0	1
0	1	1
1	1	0

AND



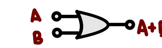
Input A	Input B	output
0	0	0
1	0	0
0	1	0
1	1	1

NOR



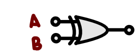
Input A	Input B	output
0	0	1
1	0	0
0	1	0
1	1	0

OR



Input A	Input B	output
0	0	0
1	0	1
0	1	1
1	1	1

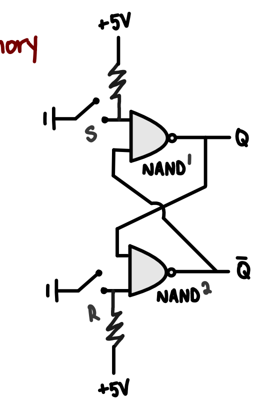
XOR



Input A	Input B	output
0	0	0
1	0	1
0	1	1
1	1	0

Bistable Memory

'Flip-Flop'



With both switches open

Guess: what if Q = 0 then $\bar{Q} = 1$ then both inputs are high V

Guess: what if Q = 1 then $\bar{Q} = 0$ then inputs of 1 are 0 or 1 V

Input S	Input R	output Q	output $\bar{Q}$
0	0		
1	0	1	0
0	1	0	1
1	1		

→ now close S then Q = 1 + $\bar{Q}$ = 0

→ now open S nothing changes!

→ now close R then Q = 1 + $\bar{Q}$ = 0

→ now open R nothing changes

Input A	Input B	output
0	0	1
1	0	1
0	1	1
1	1	0